

# TIC 250549396

Eclipsing binary candidate dossier - Star Vetter

CLEAN

## BLS EPHEMERIS

Period 2.192896 d  
Epoch (t0) 1469.1898 BTJD  
Duration 0.1700 d  
Depth 6.895 % (68,951 ppm)  
SNR 837.8

## INDEPENDENT CHECKS (f3i gates)

Eclipses observed 9 (gate  $\geq 3$ )  
Robust depth ratio 1.06 (gate  $\geq 0.50$ )  
Robust depth sigma 107.7 (gate  $\geq 7.0$ )  
Odd/even z-score 2.51 (consistent)  
Secondary eclipse not detected

## CATALOG CROSS-MATCH (novelty)

|                              |          |
|------------------------------|----------|
| VSX                          | no match |
| ASAS-SN                      | no match |
| Gaia eclipsing-binary table  | no match |
| Gaia variable classification | no match |
| TESS EB catalog (Prsa+2022)  | no match |
| ExoFOP TOI list              | no match |

Verdict: NOVEL

## PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 107.2 sigma t0 refined by 0.0025 phase  
Difference-image peak SNR 41.15 (gate  $\geq 5.0$ ) Compactness 0.85 (gate  $\geq 0.50$ )  
Dimming centroid offset 0.03 px (1 arcsec) (gate  $\leq 1.0$  px, i.e. on the target)  
Nearest bright-enough neighbour in cutout: TIC 250549395  
1.73 px away, dTmag +2.30 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

